

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT2 — DATA ANALYSIS PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Analysis (BL4)	Assessment:	Assessment Tool 2 – Data Analysis Project
Weightage:	40% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Analyze real datasets using ECDFs, skewed-distribution techniques, Q–Q plots, multi-distribution comparisons, and proportion visualizations to identify patterns and justify conclusions. (BL4)</i>		
Student Name:	M.SAI HARSHITHA	Reg. No.:	192424408
Section / Batch:	B.TECH AI & DS	Date:	13.08.2026

Code of Conduct

I M.SAI HARSHITHA/192424408 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond construction: analyze the given data, identify the relevant pattern, and justify your conclusion.
- Show all supporting calculations (e.g., IQR, percentiles, angle calculations) where relevant.
- Where a question asks you to compare or recommend, state your reasoning explicitly — a correct chart choice without justification will not receive full marks.

Annexure A – DATA ANALYSIS PROJECT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Data Analysis Project

CO3 · BT4: Analyze ·

This rubric is used to assess student responses to the CO3-AT2 Data Analysis Project, in which students analyze a given dataset using ECDF, skewed-distribution, Q–Q plot, multi-distribution, or proportion-visualization techniques and justify their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant technique (ECDF, skewed-distribution/log transform, Q–Q plot, multi-distribution comparison, or proportion visualization) and correctly identifies which pattern applies to the given data.	Good understanding of the relevant technique, with only minor gaps in identifying the correct pattern.	Basic understanding, but with some misconceptions about which technique or pattern applies to the data.	Poor understanding of core concepts; misidentifies the pattern or technique required by the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, going beyond description to explain why the pattern occurs or what it implies.	Provides clear analysis with adequate reasoning tied to the data, for most parts.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Application	20%	Effectively applies analytical techniques to compare, justify, or recommend a conclusion that is well-supported by the data and any calculations shown.	Applies analytical techniques correctly, with only minor errors in reasoning or calculation.	Limited application; conclusions are only partially supported by the data or calculations.	Fails to apply appropriate analytical techniques; conclusions are unsupported or incorrect.
Organization	15%	The response is clearly structured, addressing every sub-part (a–b) in a logical sequence with coherent overall flow.	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — pattern identification, calculations, and conclusions — are completely accurate and well-suited to the data.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

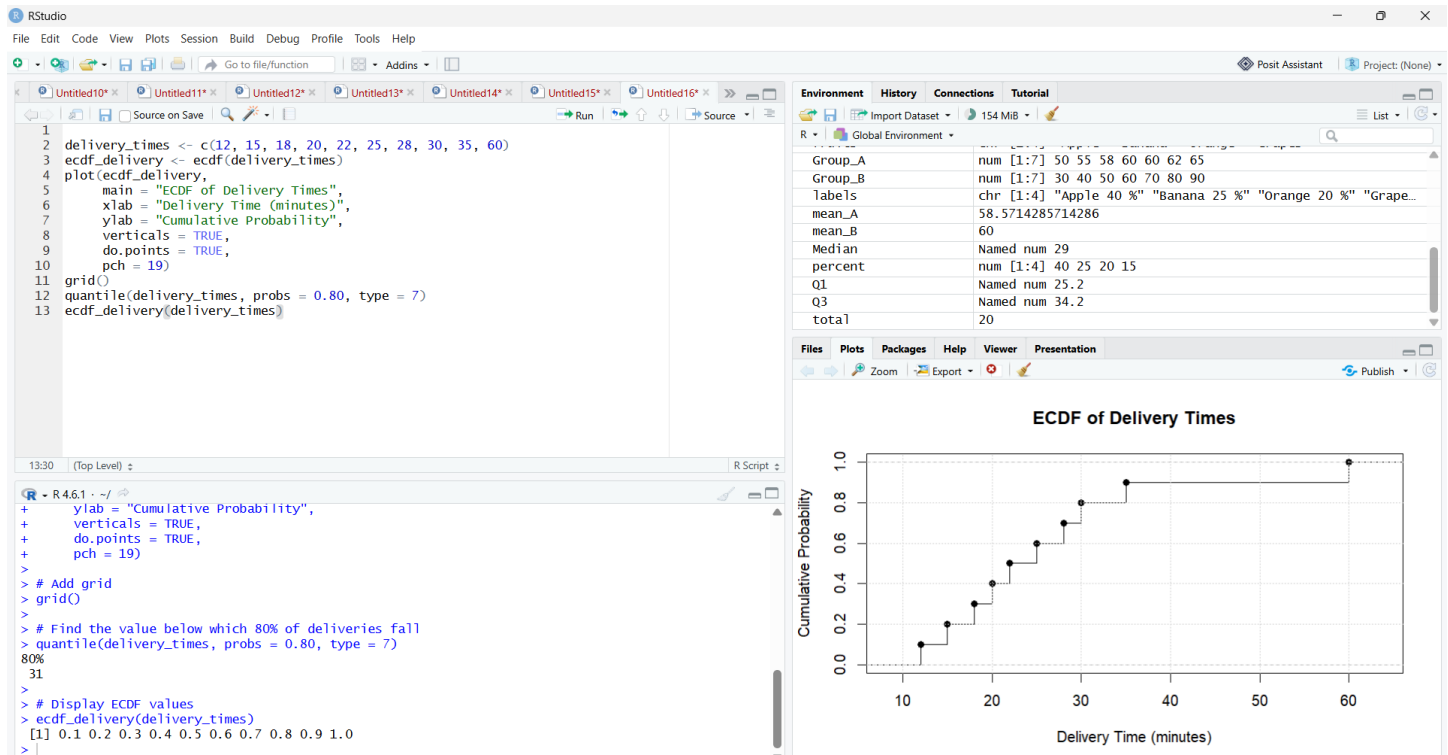
Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ECDF

Problem 1 (5 Marks)

Data: Delivery times (minutes) for 10 orders: 12, 15, 18, 20, 22, 25, 28, 30, 35, 60.

- Construct the ECDF for this data and identify the value below which 80% of deliveries fall.
- Analyze whether the ECDF suggests the data is skewed, and justify your answer using the shape of the curve.



a) ECDF Analysis

The ECDF was constructed for the delivery times using the `ecdf()` function in R. The curve reaches a cumulative probability of **0.80 at 30 minutes**. Therefore, **80% of the deliveries are completed in 30 minutes or less**.

b) Skewness Analysis

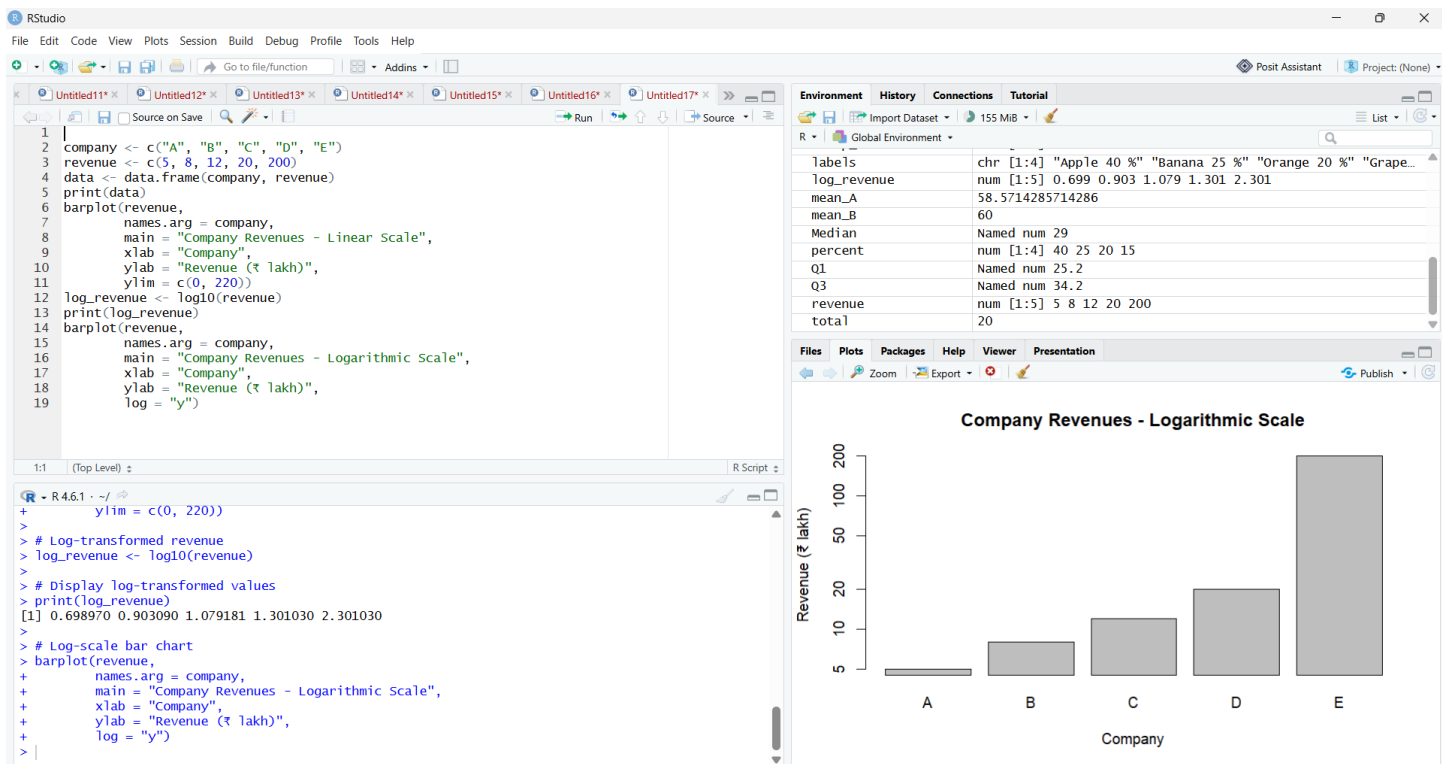
The ECDF indicates that the data is **positively skewed (right-skewed)**. Most delivery times are concentrated between **12 and 35 minutes**, while the value **60 minutes** is considerably larger than the other observations. This produces a longer tail toward the right side of the distribution, indicating the presence of a high delivery-time observation and right skewness.

Skewed Distributions & Log Transform

Problem 7 (5 Marks)

Data: Company revenues (₹ lakh): Company A = 5, B = 8, C = 12, D = 20, E = 200.

- Analyze why plotting this data on a linear-scale bar chart would be misleading.
- Recommend an alternative axis scale and explain how it resolves the problem.



a) Analysis of Linear-Scale Bar Chart

The company revenue data is highly positively skewed because Company E has a revenue of **₹200 lakh**, while the remaining companies have revenues between **₹5 lakh and ₹20 lakh**. On a linear-scale bar chart, the very large value of Company E dominates the graph. Consequently, the bars representing Companies A, B, C, and D appear very small, making their differences difficult to visualize and compare.

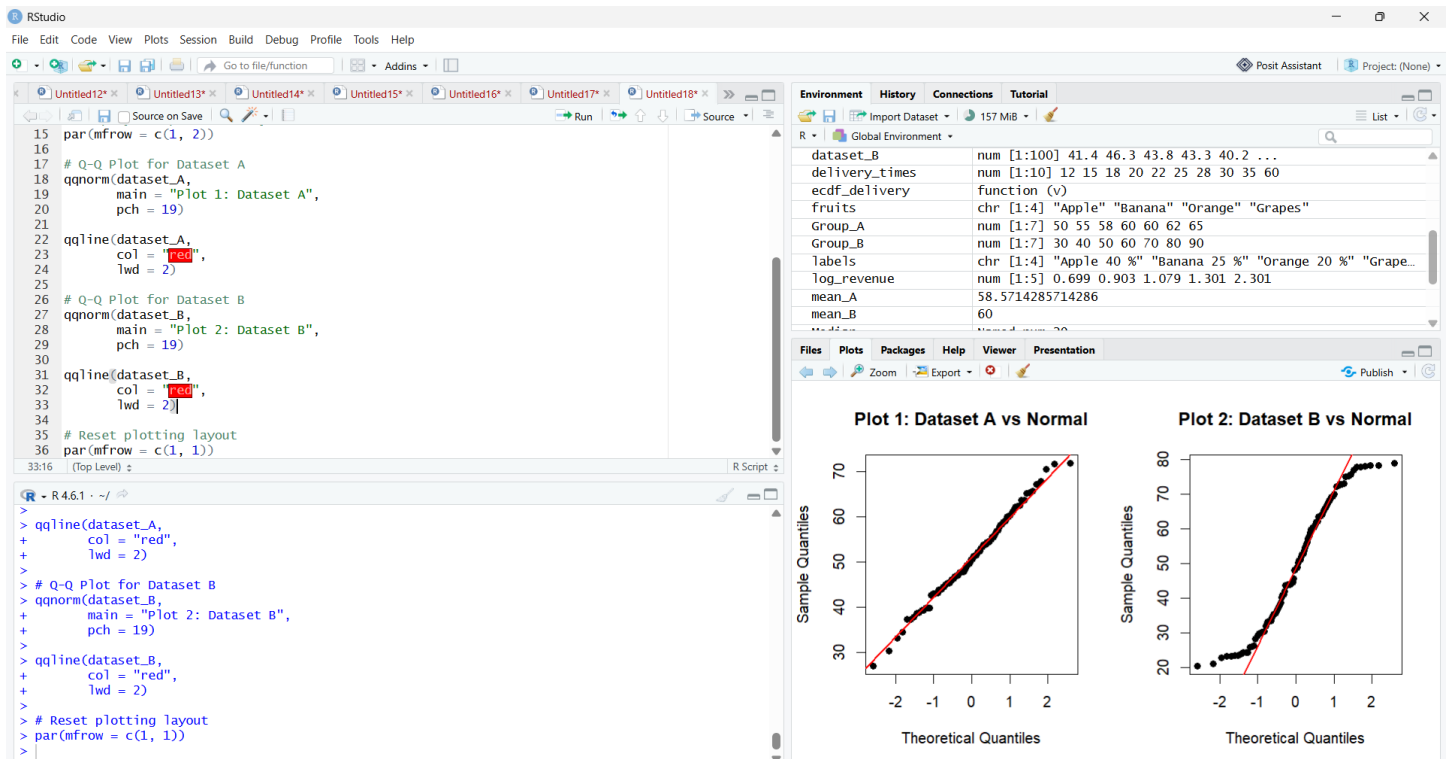
b) Alternative Axis Scale

A **logarithmic scale** on the y-axis is recommended. A log scale compresses the effect of the extreme value of ₹200 lakh while providing more visual separation among the smaller revenue values. Therefore, Companies A–D can be compared more effectively while Company E remains visible. A **log transformation such as $\log_{10}(\text{revenue})$** can also be used to reduce the influence of the extreme value and make the distribution easier to visualize.

Problem 13 (5 Marks)

Data: Two Q–Q plots are compared: Plot 1 (Dataset A vs. Normal) is a straight line; Plot 2 (Dataset B vs. Normal) shows an S-shaped curve at both ends.

- Analyze and compare what each plot tells you about its dataset's relationship to normality.
- Which dataset would you trust more for a statistical method that assumes normality? Justify your answer.



a) Analysis and Comparison

Plot 1 (Dataset A vs. Normal):

The points in Plot 1 lie approximately along a straight reference line. This indicates that the observed quantiles of Dataset A are close to the theoretical normal quantiles. Therefore, Dataset A can be considered approximately normally distributed.

Plot 2 (Dataset B vs. Normal):

Plot 2 shows an S-shaped pattern, particularly at both ends of the Q–Q plot. This systematic departure from the straight reference line indicates that Dataset B does not follow a normal distribution. The deviations at the tails suggest that the extreme values differ from what would be expected under normality.

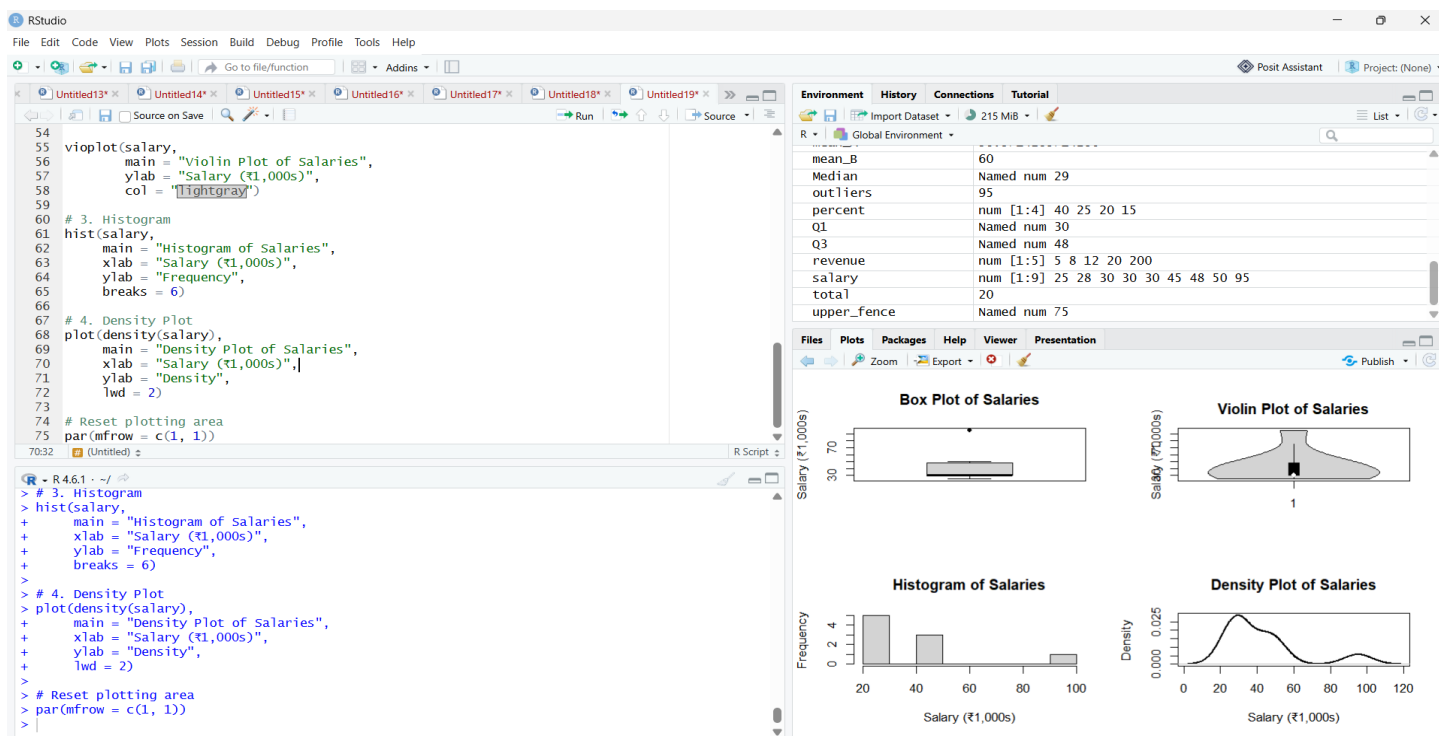
b) Dataset preferred for a normality-based statistical method

Dataset A would be trusted more for a statistical method that assumes normality. Its Q–Q plot is approximately linear, providing visual evidence that the data follows a normal distribution. Dataset B shows an S-shaped deviation from the reference line, indicating non-normality. Therefore, methods requiring normality are more appropriate for Dataset A.

Problem 19 (5 Marks)

Data: Salaries (₹1,000s) for 9 employees: 25, 28, 30, 30, 30, 45, 48, 50, 95.

- Analyze this dataset for outliers using the $1.5 \times \text{IQR}$ rule, showing your calculation.
- Explain how the presence of this outlier would affect a violin plot's shape compared to a box plot's five-number summary.



a) Outlier Analysis Using $1.5 \times \text{IQR}$ Rule

The salary data is:

25, 28, 30, 30, 30, 45, 48, 50, 95

Using R's default quartile calculation:

- Q1 = 30
- Q3 = 48
- IQR = $48 - 30 = 18$

Lower fence:

$$30 - 1.5(18) = 3$$

Upper fence:

$$48 + 1.5(18) = 75$$

Any observation below 3 or above 75 is considered an outlier. Since $95 > 75$, the salary of ₹95,000 is an outlier.

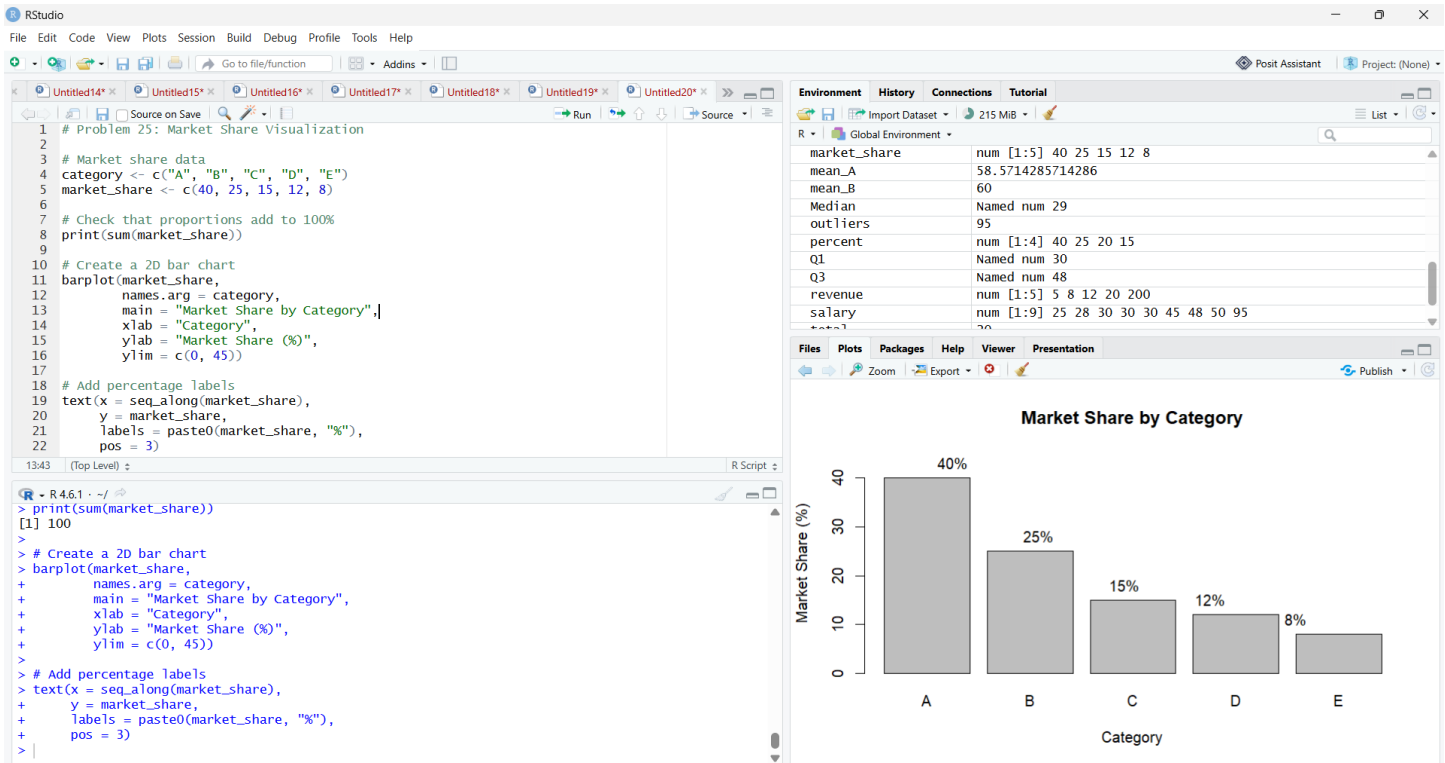
b) Effect on Violin Plot Compared with Box Plot

The outlier of ₹95,000 causes the salary distribution to become **positively skewed (right-skewed)**, producing a longer tail toward the higher salary values. In a **box plot**, the value 95 is clearly displayed as a separate point beyond the upper whisker, making the outlier easy to identify. In a **violin plot**, the outlier can extend the density toward the right and create a long tail. However, because the dataset contains only **9 observations**, the violin plot's estimated density should not be over-interpreted. The box plot is more reliable for identifying the outlier using the $1.5 \times \text{IQR}$ rule.

Problem 25 (5 Marks)

Data: A report uses a 3D pie chart, tilted at an angle, to show 5 categories' market share.

- Analyze the specific visual distortion this chart introduces and explain how it could lead a reader to misjudge the data.
- Propose an alternative chart type and explain why it avoids the distortion you identified.



a) Analysis of Visual Distortion

A tilted 3D pie chart can introduce **perspective distortion**. Slices positioned toward the front may appear larger, while slices toward the back may appear smaller, even though their actual market shares remain unchanged. The 3D depth and tilt therefore distort the perceived size of the sectors. This can cause readers to **overestimate the proportion of some categories and underestimate others**, resulting in a misleading interpretation of the market-share data.

b) Alternative Chart

A **2D bar chart** is recommended as an alternative. In a bar chart, the length or height of each bar directly represents the market share, and all categories are displayed using the same scale and baseline. It does not introduce perspective or artificial depth, so the actual proportions can be compared more accurately. A simple 2D pie chart is also better than a 3D pie chart because it removes perspective distortion.